Nagamitesh Nagamuralee

PS #1

Date submitted: 2/3/26
Due : 2/3/26

Acknowledgement:
Compared solutions with Pranav Shankar

Nagamitesh Nagamurake

Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

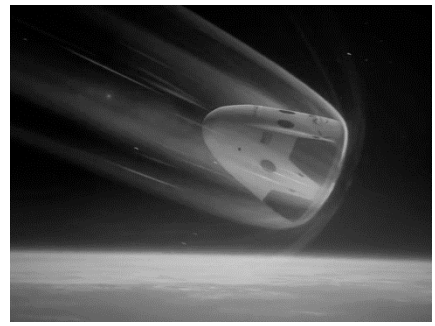
Spring 2026

Problem Set #1

Posted: Monday, Jan 26 **Due: Tuesday, Feb 3, 11:59 pm**

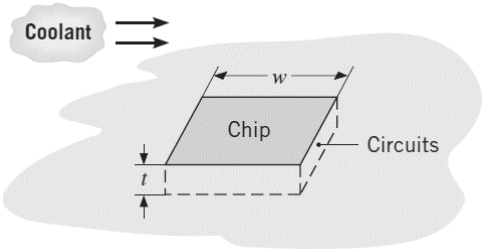
---

1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.



Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

2. (30 pt) A square silicon chip (k = 150 W/(m·K)) is of width w = 5 mm on a side and of thickness t = 1 mm. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is T_∞ = 20 °C.



(a) What is the steady-state temperature difference between back and front surfaces of the chip?

(b) If the front surface temperature of the chip cannot be above 80 °C, calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

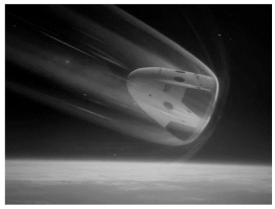
3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha_s = 0.76$.

(a) Consider a day of solar irradiation $G_s = 750$ W/m^2, ambient temperature $T_\infty = 300$ K, surrounding temperature $T_{surr} = 300$ K, and heat transfer coefficient of air flow $h = 10$ W/(m$^2\cdot$ K), calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Assume the roof can be treated as a grey body to the irradiation of surrounding environment. Would the neighboring cat be comfortable walking on the roof constructed by this material?

(b) According to your calculation, discuss the dominant heat transfer mode in (a).

1

1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.



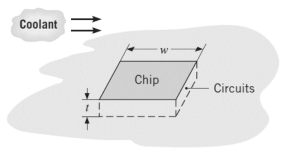
Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

Outside the atmosphere there is no air & thus no heating due to conduction since no matter to cause frizional heat or conduct. The lack of any fluid also means no convection, thus dominant heat transfer mode will be radiation from the sun. Any uneven heating can also cause conduction through the body of the craft.

During reentry the main mode will be conduction because due to the high speed reentry the travel through the air will create a lot of friction & that heat will be conducted through the shell of the spacecraft. Due to the atmosphere, convection will also be a large mode of heat transfer.

②

2. (30 pt) A square silicon chip (k = 150 W/(m·K)) is of width w = 5 mm on a side and of thickness t = 1 mm. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is T_∞ = 20 °C.



(a) What is the steady-state temperature difference between back and front surfaces of the chip?

(b) If the front surface temperature of the chip cannot be above 80 °C, calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

Known:

- $k = 150 \frac{W}{m \cdot k}$
- $W = 5$ mm → width
- $t = 1$ mm → thickness
- sides and back is thermally insulated (no conduction)
- Front has coolant with convection, 9 W dissipated at steady state
- $T_{coolant} = 20°C$

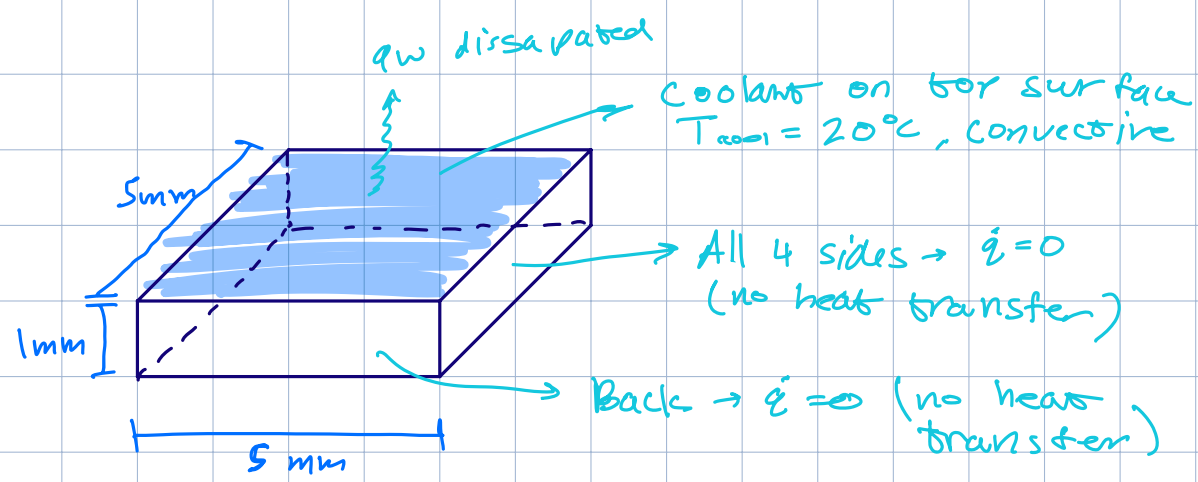
Find

a) Steady state temp. diff. between back & front of chip, ΔT

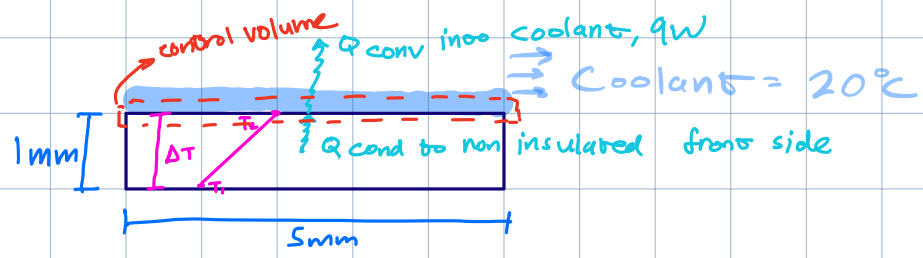
b) Chip front temp not more than 80°C, find minimum heat transfer coeff for coolant (also compare with book and find conv. processes)

Schematic:

Trimetric View:



Side View:



Assumptions: · Assuming that no thermal radiation
· Assuming potential & kinetic energy changes are negligible
· At steady state, so no net heat transfer
· no heat transfer through insulated sides

Analysis:

a) Doing an energy balance (only on the surface of front) to find the steady state ΔT from the front to back

$$\dot{E}_{in} + \cancel{\dot{E}_{generated}} - \dot{E}_{out} = \cancel{\dot{E}_{steady}}$$

no generated

no change in steady state so 0

$\dot{E}_{in}$ as in schematic $= q_{cond}$

$\dot{E}_{out}$ as in schematic $= q_{conv}$

$$q_{cond} - q_{conv} = 0$$

$$-k\underset{w^2}{A}\frac{dT}{dx} - q_{conv} = 0$$

$$q_{conv} = -kw^2\frac{dT}{dx} = -kw^2\frac{\Delta T}{t}$$

$$\frac{(q_{conv})(t)}{(-k)(w^2)} = \Delta T \rightarrow \Delta T = \frac{(9\,W)(0.001\,m)}{(-150\,\frac{W}{m\cdot k})(0.005\,m)^2}$$

$$\Delta T = -2.4\,k$$

so temperature difference $= 2.4\,k$

b) Here finding the minimum heat transfer coefficient to keep surface at 80°c

Using newton's law of cooling to find h

$$q_{conv} = hA(T_s - T_\infty)$$

$$h \geq \frac{q_{conv}}{A(T_s - T_\infty)} = \frac{q_w}{(0.005\ m)^2 (80 - 20\ K)}$$

$$h \geq 6000\ \frac{W}{m^2 \cdot K}$$

The convective heat transfer coefficient value can be reached by the forced convection of a liquid as seen in textbook table 1.1 since it has a range of 100 – 20000.

3

3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha_s = 0.76$.

(a) Consider a day of solar irradiation G_s = 750 W/m^2, ambient temperature T_∞ = 300 K, surrounding temperature T_{surr} = 300 K, and heat transfer coefficient of air flow h = 10 W/(m^2· K), calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Assume the roof can be treated as a grey body to the irradiation of surrounding environment. Would the neighboring cat be comfortable walking on the roof constructed by this material?

(b) According to your calculation, discuss the dominant heat transfer mode in (a).

Known:

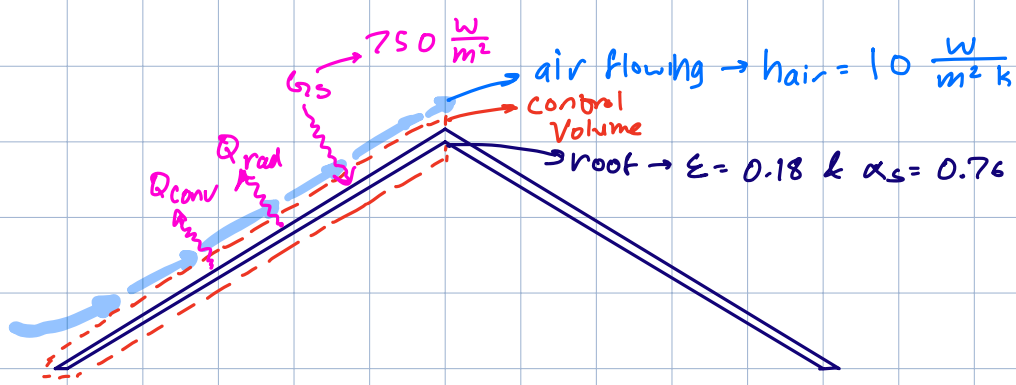
- Roof material: $\varepsilon = 0.18$ and $\alpha_s = 0.76$ for solar radiation
- Solar irradiation, $G_s = 750$ W/m^2
- $T_\infty = 300$ K (ambient)
- $T_{surr} = 300$ K (surrounding)
- heat transfer coeff of air flow $= 10 \frac{W}{m^2 \cdot K}$
- Bottom of roof is thermally insulated

Find:

a) would cat be comfy (roof temp) walking on roof material?

b) what is dominant heat transfer mode?

Schematic:



Assumption:
- roof is a grey body for radiation
- The control volume is infitessmly thin
- Assuming no other mode of energy transfer

Analysis:

a) Finding temp of roof to find if cat will be comfy so doing a steady-state energy balance

$$\dot{E}_{in} = \dot{E}_{out}$$

$$q_{rad} = q_{conv} + q_{rad}$$

$$\alpha_s G_s = h(T_{sfc} - T_\infty) + \varepsilon\sigma(T_{sfc}^4 - T_{surr}^4)$$

$$\alpha_s G_s = hT_{sfc} - hT_\infty + \varepsilon\sigma T_{sfc}^4 - \varepsilon\sigma T_{surr}^4$$

$$\alpha_s G_s + hT_\infty + \varepsilon\sigma T_{surr}^4 = hT_{sfc} + \varepsilon\sigma T_{sfc}^4$$

$$10\,T_{sfc} + (0.1)(5.67\times10^{-8})T_{sfc}^4 = (0.76)(750) + (10)(300) + (0.18)(5.67\times10^{-8})(300)^4$$

Then solving with online calculator finding that $T_{sfc} = 349.96\ K = 76.8°C$

That is too hot to walk on roof since that is close to boiling & even a 100 °F day or temp is hot & thats almost double that so certainly too hot for cat.

b) Computing the radiation & convection leaving roof to figure out most dominant

$$q_{rad} = \varepsilon\sigma T_{sfc}^4 = 0.18(5.67\times10^{-8})(350)^4 = 153.2\ W/m^2$$

$$q_{conv} = h(T_{sfc} - T_{surr}) = 10(350-300) = 500\ W/m^2$$

Thus dominant mode is convection since it transfers $500\ \frac{W}{m^2}$ where as conduction only $153.2\ W/m^2$